Answer Key & Teacher Notes

Session 1 — Writing Inequalities · Session 2 — Representing Inequalities

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Writing Inequalities

TALK PROMPTS

- Why can an inequality have many solutions when an equation usually has one?
- Why does the phrase "at least" let the limit number itself count as a solution?

PARTNER TASK

Movie Age Rule and Science Fair Ribbon. One partner writes $a \geq 13$ for a film rule; the other writes $p > 85$ for a ribbon. Each names two solutions and one non-solution, then explains whether the boundary number counts.

WATCH FOR

- Using > for "at least," which leaves out the boundary number.
- Reversing the symbol ($x \leq 20$ for "at least 20").
- Thinking "at most 10" and "less than 10" mean the same thing.

SESSION 2 · TEACHER NOTES

Representing Inequalities

TALK PROMPTS

- How does a number line show that an inequality has many solutions?
- How does the kind of circle you draw change the list of lengths that fit?

PARTNER TASK

Form A, problems 5 and 6. One partner writes $s \leq 5$ for Lab Station and graphs it; the other names two solutions and one non-solution and checks them on the graph. Switch roles with Parade Float ($h < 14$).

WATCH FOR

- Using a closed circle for < or > (the boundary is not a solution).
- Drawing the arrow the wrong way, such as pointing right for ≤.
- Ignoring the story, such as listing 2.5 people as a solution.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.B.8	1	2	2	2	2	2	2	3	2	1	1
S1 · Form B	6.EE.B.8	1	2	2	2	2	2	2	3	2	1	1
S2 · Form A	6.EE.B.8	1	2	2	2	2	2	2	3	2	1	2
S2 · Form B	6.EE.B.8	1	2	2	2	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

inequality: A math sentence that **compares** using <, >, ≤ or ≥.

at least: That number **or more** (≥).

at most: That number or **less** (≤).

more than: **Greater** than the number, not equal to it (>).

- First decide whether the boundary number is **allowed**.
- Allowed: use ≤ or **≥**. Not allowed: use < or **>**.

Example: Key words: "**more than**." Is 20 allowed? **No**, so use >. Inequality: L **>** 20. Answer: $L > 20$, so 20 laps is **not** a solution.

YOU TRY

1. a. $x \geq 13$ b. $x < 9$

2. $d \leq 50$; yes, $50 \leq 50$ is true.

EXIT TICKET

1. a. $x \leq 40$ b. $x > 6$

2. $c > 300$; no, $300 > 300$ is false

3. **C** (at least)

PRACTICE FORM A

1. a. $x \geq 18$ b. $x \leq 7$ c. $x > 25$ d. $x < 12$ e. $x \leq 15$ f. $x \geq 4$

2. a. $a \geq 13$ b. $p > 85$ c. $h \geq 30$ d. $w \leq 15$

3. $t \geq 150$; sample solutions: 150 and 200

4. $g > 9$; no, $9 > 9$ is false

5. $c \geq 4$; no, $3 \geq 4$ is false

6. $s \geq 25$; sample solutions: 25 and 40

7. a. $p \leq 12$ b. Yes, because $12 \leq 12$ is true.

8. a. $h \geq 42$ b. $h \leq 76$ c. Sample: 50 and 60 fit; 80 does not.

9. $h > 48$ leaves out 48, but a rider who is exactly 48 inches tall may ride. "At least" includes 48: $h \geq 48$.

10. **B** ($x \geq 20$)

11. **B** ($x \leq 9$)

Self-check word **SYMBOL**

PRACTICE FORM B

1. a. $x \geq 21$ b. $x \leq 6$ c. $x > 40$ d. $x < 10$ e. $x \leq 50$ f. $x \geq 3$

2. a. $h > 48$ b. $c \leq 6$ c. $m \geq 10$ d. $t < 5$

3. $t \geq 200$; sample solutions: 200 and 250

4. $g > 12$; no, $12 > 12$ is false

5. $c \geq 5$; no, $4 \geq 5$ is false

6. $s \geq 30$; sample solutions: 30 and 45

7. a. $p \leq 15$ b. Yes, because $15 \leq 15$ is true.

8. a. $h \geq 44$ b. $h \leq 78$ c. Sample: 50 and 60 fit; 40 does not.

9. ≥ includes 300, but exactly 300 cans is not more than 300. The boundary is not allowed: $c > 300$.

10. **D** ($x > 30$)

11. **A** ($x \leq 9$)

Self-check word **LIMITS**

SESSION 2 · ANSWERS

GUIDED NOTES

open circle: The boundary is **not** a solution (< or >).

closed circle: The boundary **is** a solution (≤ or ≥).

solutions of an inequality: All the numbers that make it true.

- Draw a circle at the **boundary** number.
- For < or ≤, the arrow points **left**. For > or ≥, it points **right**.

Example: The boundary is **3**. < does not include 3, so draw an **open** circle. Draw the arrow to the **left**. Answer: Sample solutions: 2, 0, −1.5

YOU TRY

1. $x \geq -2$

2. a. no b. yes

EXIT TICKET

1. $x \leq 1$

2. a. yes b. no

3. **A** (open circle at 3, arrow right)

PRACTICE FORM A

1. a. $x > 4$ b. $x \leq -1$ c. $x \geq 6$ d. $x < 0$

2. a. $n \leq 120$ b. closed circle at 120, arrow left

3. $t < 78$; no, the open circle shows 78 is not a solution

4. $h < 22$; sample: 20 and 21.5 are solutions; 22 is not

5. $s \leq 5$; yes, $5 \leq 5$ is true (whole numbers 0 to 5)

6. a. $h < 14$ b. open circle at 14, arrow left

7. Sample: 8, 9.5 and 20 are solutions; 7 is not (any number 7 or less is not).

8. a. $w \geq 50$ and $w \leq 900$ b. Sample: 50, 400 and 900 c. "At most" includes 900 ($900 \leq 900$ is true), but $901 \leq 900$ is false.

9. < leaves out 6, but carrying exactly 6 items is allowed. "No more than" includes 6: $i \leq 6$ (closed circle at 6).

10. **D** (6)

11. **B** (3.5)

Self-check word **CIRCLE**

PRACTICE FORM B

1. a. $x < 2$ b. $x \geq -3$ c. $x \leq 5$ d. $x > -1$

2. a. $n \leq 90$ b. closed circle at 90, arrow left

3. $t < 80$; no, the open circle shows 80 is not a solution

4. $h < 24$; sample: 20 and 23.5 are solutions; 24 is not

5. $s \leq 6$; yes, $6 \leq 6$ is true (whole numbers 0 to 6)

6. a. $h < 13$ b. open circle at 13, arrow left

7. Sample: 5, 6.5 and 12 are solutions; 4 is not (any number 4 or less is not).

8. a. $w \geq 40$ and $w \leq 800$ b. Sample: 40, 300 and 800 c. "At most" includes 800 ($800 \leq 800$ is true), but $801 \leq 800$ is false.

9. < leaves out 32, but seating exactly 32 students is allowed. "At most" includes 32: $s \leq 32$ (closed circle at 32).

10. **A** (9)

11. **D** (7.5)

Self-check word **ARROWS**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · WRITING INEQUALITIES

1. a. $x > 12$ b. $x \leq 30$ c. $x < 8$ d. $x \geq 15$ e. $x \leq 6$ f. $x > -3$ g. $x < 0$ h. $x \leq 100$

2. a. $h \geq 54$ b. $s \leq 35$ c. $n > 10$ d. $p < 120$

3. $r \leq 48$. Yes, 48 riders is allowed because $48 \leq 48$ is true. No, 50 riders is not allowed because $50 \leq 48$ is false.

4. Sample: A movie must be shorter than 60 minutes ($t < 60$). Solutions: 45 and 59.5 (any number less than 60).

5. No. "At most 10" is $x \leq 10$, so 10 is a solution. "Less than 10" is $x < 10$, so 10 is not a solution.

SHEET 2 · REPRESENTING INEQUALITIES

1. $x > -1$

2. Closed dot at 2, arrow right. Sample solutions: 2 and 5 (any number 2 or greater).

3. a. $x > 3$ b. $x \leq -2$

4. $w \leq 24$. Closed dot at 24, arrow left.

5. The dot should be open. > does not include the boundary: $5 > 5$ is false, so 5 is not a solution. (His arrow to the right is correct.)

6. **B** ($x \leq 4$)

7. **D** (0)

October 2026 conceptual additions — teacher key

Session 1 · Writing Inequalities

h≥120. 119 fails; 120 and 121 satisfy. At least includes equality.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: h≥120. 119 fails; 120 and 121 satisfy. At least includes equality.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Representing Inequalities

m>5; open boundary at 5, ray right. Actual feasible member counts are 6,7,8,...; 5.5 is numerically above 5 but not a whole-person count.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: m>5; open boundary at 5, ray right. Actual feasible member counts are 6,7,8,...; 5.5 is numerically above 5 but not a whole-person count.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.